
Annex 6 — Sector: Airport Waste — CDG (France)

Purpose: Sector-specific feedstock profile for Paris-Charles de Gaulle Airport waste processed through the Urban Rig, including step-by-step derivation of expected pyrolysis product yields (oil, off-gas, residue).

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OBJECTIVE

To answer the question: **If we source waste from CDG Airport operations, what composition enters the Urban Rig, and what are the expected outputs?**

CDG Airport generates a fundamentally different waste stream from French municipal waste (Annex 5). Airport waste is richer in plastics, packaging, and food service materials, with less putrescible garden waste and fewer sanitary textiles. This shifts the yield profile significantly toward higher oil output and lower char production.

This annex follows the same Part 1 / Part 2 structure as Annex 5. Every parameter is classified as: **FACT** (measured or from official sources), **HYPOTHESIS** (literature-based estimate), or **ASSUMPTION** (engineering judgment).

PART 1 — WHAT GOES IN: Airport Waste Composition

This part answers: of the waste available at CDG, what materials would enter the reactor, and in what proportions?

1. CDG AIRPORT — OVERVIEW AND WASTE VOLUMES

1.1 Airport Profile

Parameter	Value	Source
Passengers (2024)	70.3 million	Groupe ADP traffic figures, Feb 2025
Passengers (2023)	68.3 million	Groupe ADP traffic figures
Terminals	3 (T1, T2A-F, T3)	Groupe ADP
Cargo zone	Paris-CDG Cargo City (2nd largest in Europe)	Groupe ADP
Waste contractor	Paprec (since Jan 2016)	Paprec Solutions / Industrial News (L'Usine Nouvelle), 2015

[FACT] **FACT** — Traffic figures from official Groupe ADP publications. Data quality: HIGH.

1.2 Total Waste Generation

Parameter	Value	Source
Total non-hazardous waste (all Paris airports)	~43,000 T/year	Paprec Solutions
CDG share (estimated)	~35,000 T/year	ASSUMPTION — CDG handles ~80% of Groupe ADP (Paris Airports Group) passenger + cargo traffic
Waste per passenger (terminal)	~0.4–0.5 kg/pax	Istanbul Airport study (Ozbay & Gokceviz, 2021): 0.39 kg/pax (2019). Cross-check: Heathrow ~1.4 kg/pax includes cabin waste.
Cabin waste per passenger (IATA)	0.94 kg/pax (international average)	IATA/ASF Cabin Waste Composition Audits, 2024

[HYPOTHESIS] **ASSUMPTION** — CDG-specific waste tonnage is not publicly broken out from Groupe ADP totals. The 35,000 T estimate is derived from CDG's ~80% share of Groupe ADP traffic. Actual tonnage may differ due to cargo zone contribution.

Note on waste destination today: Of the ~43,000 T/year across Paris airports, Paprec reports that ~27,000 T are incinerated and only ~5,000 T recycled (source: Paprec Solutions). The valorization rate is ~50%, with material recovery around 12%. This represents significant untapped potential.

2. AIRPORT WASTE STREAMS — SOURCE IDENTIFICATION

CDG waste comes from five distinct operational zones, each with a different composition profile:

Stream	Description	Est. share of CDG waste	Key materials
S1: Terminal commercial	Shops, restaurants, lounges, passenger areas	~30%	Food waste, plastic packaging, cups, paper, cardboard, glass
S2: Cabin/catering waste	Inflight meal trays, beverages, service items	~25%	Food waste, plastic trays/cups, aluminium foil, paper napkins, glass miniatures
S3: Cargo/freight zone	Packaging from air freight operations	~25%	Cardboard, shrink wrap (LDPE), wood pallets, plastic film, strapping
S4: Offices/admin	Airport offices, airline offices, ground handlers	~10%	Paper, cardboard, plastic bottles, food packaging
S5: Maintenance/operations	Technical areas, construction, maintenance	~10%	Mixed (some hazardous — excluded), wood, plastic, metal

[HYPOTHESIS] **ASSUMPTION** — Stream percentages are engineering estimates based on airport waste literature (Istanbul Airport study; ICAO Waste Management at Airports toolkit; Ferrovial airport waste analysis). CDG-specific breakdown not published. MEDIUM confidence.

Critical regulatory note — International Catering Waste (ICW): Cabin waste from flights originating outside the EU is classified as **Category 1 (CAT1) animal by-product** under EU Regulation 1069/2009. This waste **MUST** be disposed of by incineration, pressure sterilisation, or authorised deep burial. It **CANNOT** be recycled or composted under current EU law. This represents a significant portion of S2 at CDG (an international hub). **Pyrolysis at >400°C would meet the thermal destruction requirement, but regulatory approval would be needed.** This is flagged as a critical regulatory pathway to investigate. IATA is actively lobbying for smarter regulation of ICW (joint statement, August 2022).

3. COMPOSITION BY STREAM — DETAILED BREAKDOWN

3.1 Stream S1: Terminal Commercial Waste (~30% of total)

Source: Istanbul Airport terminal waste characterization (Ozbay & Gokceviz, 2021, *J. Material Cycles and Waste Management*); cross-validated with Schiphol BCAM study (van der Tuin-Rademaker et al., 2024, *Frontiers in Sustainability*); adjusted for CDG commercial mix.

Material	% of S1	Per 1,000 kg S1	Note
Food waste (organic)	35%	350 kg	Restaurant leftovers, passenger food scraps
Plastic packaging	20%	200 kg	Cups, bottles, trays, film, clamshells
Paper/cardboard	18%	180 kg	Newspapers, magazines, receipts, boxes

Material	% of S1	Per 1,000 kg S1	Note
Glass	8%	80 kg	Bottles (wine, spirits, water)
Metal (cans, foil)	5%	50 kg	Aluminium cans, steel food cans
Wood	3%	30 kg	Crates, pallets from retail supply
Textiles/other	5%	50 kg	Wipes, napkins, mixed
Liquids (non-processable)	6%	60 kg	Beverages, sauces — removed by drainage

[HYPOTHESIS] **HYPOTHESIS** — Based on Istanbul Airport terminal data (56% combustible, 36% recyclable) and Schiphol BCAM priority streams. Adapted for CDG’s heavy food service presence (T2E/F duty-free zones, 50+ restaurants). MEDIUM confidence.

3.2 Stream S2: Cabin/Catering Waste (~25% of total)

Source: LIFE Zero Cabin Waste project (IBERIA/Madrid Barajas, 2019); IATA Cabin Waste Handbook (2017); IATA/ASF Cabin Waste Composition Audits (2024); “Beyond the Runway” UK study (2025).

Material	% of S2	Per 1,000 kg S2	Note
Food waste (organic)	54%	540 kg	Largest fraction. 23% of total is untouched food (IATA 2024).
Plastic (trays, cups, film)	14%	140 kg	PET/PP trays, PS cups, PE film
Paper/cardboard	16%	160 kg	Napkins, cartons, meal boxes, newspapers
Glass	6%	60 kg	Miniature bottles, wine glasses
Aluminium	4%	40 kg	Foil lids, cans, meal tray covers
Liquids (non-processable)	6%	60 kg	Unconsumed beverages — drained

[FACT] **FACT** — These percentages are from the LIFE Zero Cabin Waste characterization study: organic 54%, paper/cardboard 15.6%, plastics 14.2%, glass 6.4%, aluminium 4%. Data quality: HIGH (145 flights audited at Madrid Barajas).

Caveat — France-specific: CDG serves predominantly European and long-haul routes. Long-haul flights generate more waste per passenger (0.6–0.9 kg/pax vs 0.31 kg/pax short-haul) with higher food content. The “Zero Cabin Waste” data is from Iberia at Madrid — a comparable European hub but with a different route mix.

3.3 Stream S3: Cargo/Freight Zone (~25% of total)

Source: Airport cargo packaging literature; ICAO Waste Management at Airports toolkit; Paprec CDG operational descriptions.

Material	% of S3	Per 1,000 kg S3	Note
Cardboard/corrugated	40%	400 kg	Shipping boxes, dividers, corner protectors
Plastic film (LDPE)	20%	200 kg	Shrink wrap, stretch wrap, pallet covers
Wood (pallets, crates)	25%	250 kg	Pallets (often ISPM-15 treated), crates
Strapping (PP/PET)	5%	50 kg	Banding material
Metal (banding, containers)	5%	50 kg	Steel strapping, container fragments
Other (foam, padding)	5%	50 kg	EPS foam, bubble wrap (PE), air pillows

ASSUMPTION — Based on general air freight packaging composition and operational descriptions (cardboard, wood, plastic film as primary streams). CDG-specific cargo waste characterization not published. MEDIUM confidence.

Key advantage for pyrolysis: Cargo waste is predominantly clean, dry, and plastic/cellulose-rich. Very low food contamination. The LDPE (low-density polyethylene) film + PP (polypropylene) strapping fraction is high-yield pyrolysis feedstock (77-80% oil). This is the most attractive stream for the Urban Rig.

3.4 Stream S4: Office/Admin Waste (~10% of total)

Source: Istanbul Airport office waste characterization (Ozbay & Gokceviz, 2021).

Material	% of S4	Per 1,000 kg S4	Note
Paper	35%	350 kg	Office paper, documents, envelopes
Cardboard	15%	150 kg	Delivery boxes
Plastic	15%	150 kg	Bottles, packaging, cups
Food waste	20%	200 kg	Canteen/lunch waste
Glass	5%	50 kg	
Metal	5%	50 kg	Cans
Other	5%	50 kg	

[HYPOTHESIS] **HYPOTHESIS** — Based on Istanbul Airport office data (43% recyclable, 43% combustible, 7-23% organic depending on office type). MEDIUM confidence.

3.5 Stream S5: Maintenance/Operations (~10% of total)

Material	% of S5	Per 1,000 kg S5	Note
Wood	25%	250 kg	Construction/maintenance timber
Plastic (mixed)	15%	150 kg	Piping, sheeting, packaging
Metal	20%	200 kg	Scrap, structural offcuts
Cardboard	10%	100 kg	
Inert (concrete, rubble)	20%	200 kg	Non-processable
Other/hazardous	10%	100 kg	Excluded from pyrolysis

[HYPOTHESIS] **ASSUMPTION** — Engineering estimate. LOW confidence. This stream is the least well-characterised and most variable.

4. CONSOLIDATED AIRPORT WASTE COMPOSITION

Weighting each stream by its share of CDG total waste:

Material	S1 (30%)	S2 (25%)	S3 (25%)	S4 (10%)	S5 (10%)	Weighted Total	% of total
Food/organic	105	135	0	20	0	260	26.0%
Plastic (all types)	60	35	75	15	15	200	20.0%
Paper/cardboard	54	40	100	50	10	254	25.4%
Wood	9	0	62.5	0	25	96.5	9.7%
Glass	24	15	0	5	0	44	4.4%
Metal	15	10	12.5	5	20	62.5	6.3%
Liquids (drained)	18	15	0	0	0	33	3.3%
Inert/non-processable	0	0	0	0	20	20	2.0%
Other/hazardous (excluded)	15	0	0	5	10	30	3.0%
TOTAL	300	250	250	100	100	1,000	100%

Values are kg per 1,000 kg of mixed CDG waste.

5. REMOVING NON-PROCESSABLE MATERIAL

Removed material	Mass (kg)	Removal method
Glass	44	Mechanical screening / optical sorting
Metal (ferrous + non-ferrous)	62.5	Magnetic + eddy current separation
Liquids	33	Drainage at reception
Inert (concrete, rubble)	20	Density separation
Hazardous / excluded	30	Manual sorting / pre-screening
Total removed	189.5	

Removed material	Mass (kg)	Removal method
Material entering reactor	810.5	

ASSUMPTION — Pre-sorting at standard Materials Recovery Facility (MRF) efficiency. Higher removal rate than OMR (189.5 kg vs 109.5 kg) because airport waste has more glass, metals, and liquids.

6. CONSOLIDATED EFFECTIVE FEEDSTOCK — FINAL TABLE

6.1 Four-Category View (810.5 kg entering reactor)

Category	Mass (kg)	% of reactor input	Composition
Organic	260	32.1%	Food waste from terminals, catering, offices
Cellulose	350.5	43.3%	Paper 154 + cardboard 100 + wood 96.5
Plastic-like	200	24.7%	All polymer types — see §7
Mineral inert	0	0%	All removed in pre-sort
TOTAL	810.5	100%	

6.2 Three-Category View (processable material)

Category	Mass (kg)	% of processable feedstock	Primary Urban Rig Output
Organic (food waste)	260	32.1%	Charcoal / biochar

Category	Mass (kg)	% of processable feedstock	Primary Urban Rig Output
Cellulose (paper, card-board, wood)	350.5	43.3%	Charcoal + off-gas
Plastic-like (all polymers)	200	24.7%	Fuel oil (4 fractions)
TOTAL process-able	810.5	100%	

6.3 Comparison with French OMR (Annex 5)

Parameter	French OMR (Annex 5)	CDG Airport	Difference
Plastic fraction (of processable)	23.7%	24.7%	+1.0 pp
Cellulose fraction	37.9%	43.3%	+5.4 pp
Organic fraction	38.4%	32.1%	-6.3 pp
Material removed pre-sort	109.5 kg/T	189.5 kg/T	+80 kg
Reactor input per tonne	890.5 kg	810.5 kg	-80 kg

Key insight: CDG waste is surprisingly similar to OMR in its plastic content (~25%), but significantly richer in cellulose (paper/cardboard/wood from cargo) and leaner in putrescibles (no garden waste, less food relative to packaging). The biggest difference is the pre-sort removal: more glass and metals at the airport.

7. AIRPORT PLASTIC POLYMER MIX

Airport plastic waste has a different polymer profile from municipal waste — dominated by packaging films (LDPE), PET bottles/trays, and PP containers, with very little PVC.

7.1 Estimated Polymer Distribution

Polymer	% of airport plastic	Mass in 200 kg	Source/Basis
LDPE (film, shrink wrap)	35%	70 kg	Dominant in cargo (shrink wrap) + terminal (bags)
PET (bottles, trays, clamshells)	25%	50 kg	Beverage bottles, meal trays, service items
PP (cups, lids, containers)	20%	40 kg	Food containers, cup lids, strapping
PS/EPS (foam, cups)	8%	16 kg	Styrofoam packaging, disposable cups
HDPE (bottles, containers)	7%	14 kg	Milk/juice bottles, cleaning product containers
PVC	1%	2 kg	Minimal in airport/food packaging context
Other (mixed, composites)	4%	8 kg	Multi-layer packaging, laminates

ASSUMPTION — Based on European food service and logistics packaging composition. Airport waste is packaging-dominated (not consumer goods), so PET (polyethylene terephthalate) and LDPE are overrepresented vs municipal waste. PVC (polyvinyl chloride) is minimal (food-contact packaging is PVC-free in the EU since Regulation 10/2011 restrictions). MEDIUM confidence.

Key difference from OMR (Municipal Solid Waste): PET (polyethylene terephthalate) fraction is higher (25% vs 18% in OMR) due to beverage bottles and food trays. LDPE (low-density polyethylene) is similar. PVC (polyvinyl chloride) is lower (1% vs 3%). Overall polyolefin content (LDPE+HDPE+PP) = 62% vs 64% in OMR — comparable.

PART 2 — WHAT COMES OUT: Pyrolysis Yields

Applying the same per-material yields as Annex 5 (Section 10) to the airport feed-stock.

8. WEIGHTED YIELD CALCULATION

8.1 Plastic-Like Fraction (200 kg)

Material	Mass (kg)	Oil yield	Oil (kg)	Off-gas (kg)	Residue (kg)	Data quality
LDPE	70	77%	53.9	15.4	0.7	[FACT] FACT
PET	50	45%	22.5	8.5	19.0	[HYPOTHESIS] HYPOTHE- SIS
PP	40	80%	32.0	7.2	0.8	[FACT] FACT
PS/EPS	16	58%	9.3	2.2	4.5	[FACT] FACT
HDPE	14	77%	10.8	3.1	0.1	[FACT] FACT
PVC	2	32%	0.6	0.04	0.3	[HYPOTHESIS] HYPOTHE- SIS
PVC HCl	—	—	—	1.1	—	Stoichiometric
Other	8	25%	2.0	2.0	4.0	[HYPOTHESIS] HYPOTHE- SIS
SUBTOTAL200			131.1	38.3	29.4	

Material	Mass (kg)	Oil yield	Oil (kg)	Off-gas (kg)	Residue (kg)	Data quality
Mass check			131.1 + 38.3 + 29.4 + 1.1 = 199.9 ≈ 200			
Weighted oil yield		65.6%	(131.1 / 200)			

8.2 Organic + Cellulose Fraction (610.5 kg)

Material	Mass (kg)	Oil yield	Oil (kg)	Off-gas (kg)	Residue (kg)	Data quality
Cellulose (paper, card, wood)	350.5	12%	42.1	70.1	238.3	[HYPOTHESIS] HYPOTHE-SIS
Food waste (organic)	260	10%	26.0	52.0	182.0	[HYPOTHESIS] HYPOTHE-SIS
SUBTOTAL	610.5		68.1	122.1	420.3	

8.3 TOTAL OUTPUT — 810.5 kg Entering Reactor

Product	From plastic	From organic/cellulose	TOTAL	% of 810.5 kg
Oil	131.1	68.1	199.2	24.6%
Off-gas	38.3	122.1	160.4	19.8%
Residue/Char	238.3	420.3	449.7	55.5%
HCl (filtered)	1.1	0	1.1	0.1%

8.4 Comparison with French OMR (Annex 5)

Product	French OMR	CDG Airport	Difference
Oil (% of reactor input)	23.0%	24.6%	+1.6 pp
Oil (kg per 1,000 kg raw waste)	204.7	199.2	-5.5 kg
Off-gas	19.2%	19.8%	+0.6 pp
Char/residue	57.5%	55.5%	-2.0 pp
Plastic oil yield (weighted)	65.1%	65.6%	+0.5 pp

Key finding: The oil yield per kg of reactor input is slightly higher for CDG waste (24.6% vs 23.0%), but the oil yield per tonne of raw waste is slightly lower (199.2 vs 204.7 kg) because more material is removed in pre-sorting (189.5 vs 109.5 kg). The net result: **airport waste produces roughly the same amount of oil per tonne of raw waste as OMR, but with cleaner pre-sorted recyclable by-products (glass, metals) that have independent value.**

9. ENERGY BALANCE AND CHAR ENERGY VALUE

Using the same methodology as Annex 5 §13:

Parameter	Per 1,000 kg raw CDG waste	Per day (140 T)
Combustible char	449.7 kg	62,958 kg (63.0 T)
Char energy (at 15.6 MJ/kg weighted)	7,013 MJ	981,866 MJ
Off-gas energy	~405 MJ (plastic) + ~122 MJ (organic) = ~527 MJ	73,780 MJ
Total available energy	7,540 MJ	1,055,646 MJ
Heating demand	2,714 MJ	379,960 MJ
Coverage		278%

Char energy value: The system produces ~2.8x its energy needs from char and off-gas. Under the V2 model, this char is sold commercially to fuel buyers rather

than used for reactor heating, demonstrating the fuel-grade quality of the pyrolysis char. The 278% coverage illustrates the high energy density and commercial value of the char product.

10. FINANCIAL IMPACT — CDG vs OMR

At 140 T/day throughput (steady state, Year 3):

Revenue stream	OMR (Annex 5)	CDG Airport	Note
Oil volume (L/year, full capacity)	12.56M	12.19M	Slightly less due to higher pre-sort removal
Oil revenue (€0.55/L)	€6.9M	€6.7M	
Gate fee (€/T)	€110	€130-150	Airport waste commands premium: biosecurity compliance, consistency, single-source
Gate fee revenue	€5.1M	€6.0-6.9M	Higher gate fee compensates lower oil volume

Revenue stream	OMR (Annex 5)	CDG Airport	Note
Metal revenue (€300/T)	€0.7M	€0.9M	More metal in airport waste (6.3% vs 5%)
Char revenue (V2 — all char sold)	€1.7M	€1.5M	Slightly less char surplus
Carbon credits	€2.5M	€2.5M	Similar
Glass revenue (€25/T)	€0.06M	€0.1M	More glass at airport
TOTAL REVENUE	€17.0M	€17.7-18.6M	

The CDG advantage is primarily in gate fees, not oil yield. Airport waste commands a premium because it solves a regulatory problem (ICW/CAT1 disposal) and offers consistent volume from a single counterparty (Groupe ADP / Paprec).

11. REGULATORY PATHWAY — INTERNATIONAL CATERING WASTE

11.1 Current Regulation

EU Regulation 1069/2009 classifies international catering waste (ICW) as **Category 1 animal by-product**. Permitted disposal methods:

- Incineration in approved plant
- Pressure sterilisation (133°C, 3 bar, 20 min) followed by incineration or landfill
- Deep burial in authorised landfill

11.2 Pyrolysis as Compliant Treatment

Pyrolysis at 400–600°C exceeds the thermal destruction requirements of Regulation 1069/2009. The Urban Rig operates at temperatures that would completely destroy any biological agents of concern (prions are destroyed above 600°C under prolonged exposure; bacteria and viruses are eliminated well below 400°C).

However: Pyrolysis is not explicitly listed as an approved method in the current regulation. Regulatory approval would require a petition to the competent authority (in France: DGAL — General Directorate for Food (General Directorate for Food)) demonstrating equivalent biosecurity. IATA has been lobbying for regulatory reform since 2018, with a joint statement in 2022 calling for smarter ICW regulation.

11.3 Practical Approach

Two pathways:

1. **Immediate:** Process only domestic/intra-EU cabin waste + terminal + cargo + office waste (no ICW). This represents ~60–70% of total CDG waste and avoids the CAT1 regulatory constraint entirely.
2. **Medium-term:** Seek DGAL approval for pyrolysis as equivalent thermal treatment for ICW. The thermal destruction argument is strong (>400°C vs required 133°C), but regulatory processes take 12–24 months.

12. DATA QUALITY SUMMARY

Parameter	Classification	Confidence
CDG total waste (~35,000 T/year)	[HYPOTHESIS] ASSUMPTION	MEDIUM — derived from Groupe ADP total
Stream percentages (S1-S5)	[HYPOTHESIS] ASSUMPTION	MEDIUM — airport waste literature, not CDG-specific
Terminal waste composition (S1)	[HYPOTHESIS] HYPOTHESIS	MEDIUM — Istanbul Airport study + adaptation
Cabin waste composition (S2)	[FACT] FACT	HIGH — LIFE Zero Cabin Waste project (145 flights)
Cargo waste composition (S3)	[HYPOTHESIS] ASSUMPTION	MEDIUM — packaging industry data
Office waste composition (S4)	[HYPOTHESIS] HYPOTHESIS	MEDIUM — Istanbul Airport study
Plastic polymer mix	[HYPOTHESIS] ASSUMPTION	MEDIUM — European packaging composition

Parameter	Classification	Confidence
Pyrolysis yields	Per Annex 5	See Annex 5 §16

13. RECOMMENDATIONS

1. **CDG waste is viable pyrolysis feedstock** with comparable oil yield to MSW (~24.6% vs 23.0% of reactor input) and significantly higher gate fee potential (€130–150/T vs €110/T).
 2. **The cargo zone (S3) is the single most attractive stream** — clean, dry, plastic/cellulose-rich, with minimal contamination. Isolating this stream alone (~8,750 T/year estimated) would provide high-yield feedstock at relatively low sorting cost.
 3. **Gate fee premium is the primary financial advantage**, not yield. Airport waste commands €20–40/T above municipal rates due to regulatory compliance (International Catering Waste), single-source convenience, and waste consistency.
 4. **International catering waste (Category 1 animal by-product) requires regulatory clearance** before processing. Recommend starting with non-ICW streams while pursuing DGAL (General Directorate for Food) approval for pyrolysis as equivalent thermal treatment.
 5. **A CDG-specific waste characterization study is critical.** All composition data in this annex is derived from non-CDG sources (Istanbul, Madrid, Schiphol, general literature). A 2-week audit with Paprec cooperation would provide CDG-specific data and dramatically increase confidence in the financial projections.
 6. **The metal and glass pre-sort yield at CDG is higher than MSW (Municipal Solid Waste)** (6.3% + 4.4% = 10.7% vs 5% + 5% = 10% for MSW), providing additional recyclable revenue streams.
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14. SOURCES

#	Source	Used for
1	Groupe ADP traffic figures (parisaeroport.fr, Feb 2025)	CDG passenger numbers
2	Paprec Solutions — Behind the Scenes at Paris Airports (Behind the Scenes at Paris Airports)	Waste tonnage, contractor info, energy recovery rates
3	Industrial News (Industrial News) — Paprec wins waste management contract for all Paris Airports (Paprec wins waste management contract for all Paris Airports) (2015)	Paprec contract details
4	Ozbay & Gokceviz (2021) — Towards zero-waste airports: Istanbul Airport. <i>J. Material Cycles and Waste Management</i> (PMC8527968)	Terminal/office waste composition, waste per passenger
5	van der Tuin-Rademakers et al. (2024) — Transforming waste management methods: Dutch Airport. <i>Frontiers in Sustainability</i> (10.3389/frsus.2024.1356041)	Schiphol BCAM methodology, priority streams
6	LIFE Zero Cabin Waste project (EU LIFE15 ENV/ES/000209) — Iberia/Madrid Barajas	Cabin waste composition (organic 54%, plastic 14.2%, etc.)
7	IATA/ASF Cabin Waste Composition Audits (2024)	0.94 kg/pax average, 25 flights at Changi
8	IATA — International Catering Waste: A Case for Smarter Regulation (2018)	CAT1 classification, regulatory framework
9	IATA — Sustainable Cabin fact sheet (Dec 2025)	3.6M tonnes global cabin waste, plastic 17–20%
10	IATA — Joint Statement on Smarter Regulation of ICW (Aug 2022)	Regulatory reform advocacy
11	EU Regulation 1069/2009 — Animal by-products	CAT1 classification of international catering waste

#	Source	Used for
12	ICAO — Waste Management at Airports toolkit	Airport waste categories and best practices
13	Annex 5 — MSW Feedstock Composition and Expected Yield Model	Per-material pyrolysis yields, OMR comparison

End of Annex 6